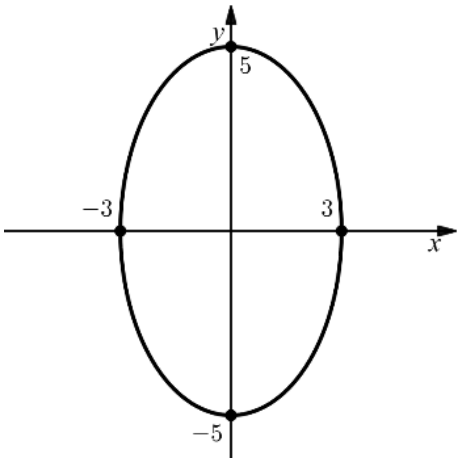
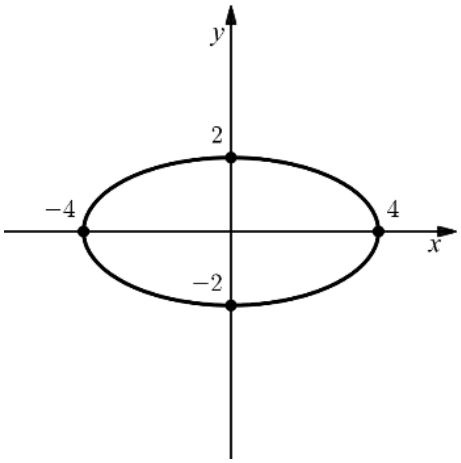
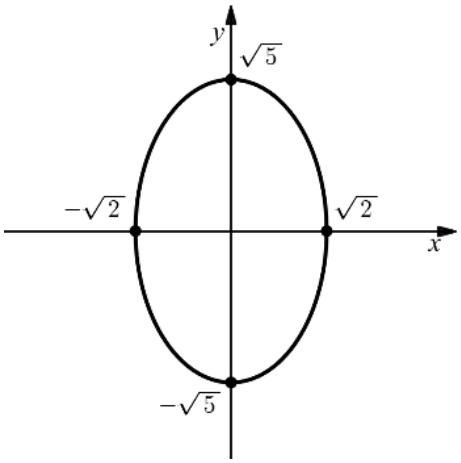
1) a)



1) b)

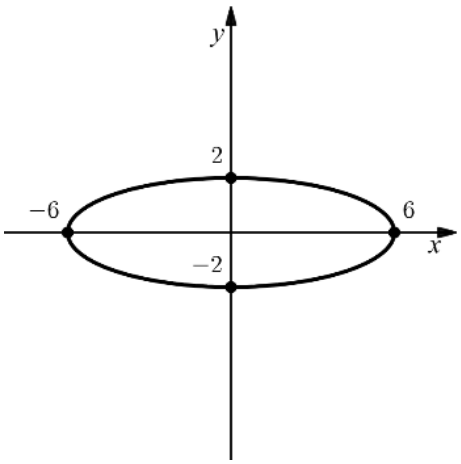


1) c)



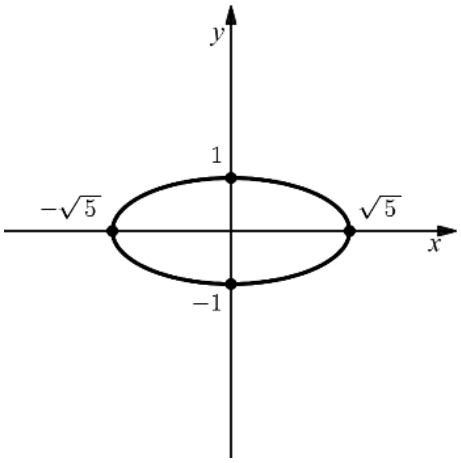
1) d)

$\frac{x^2}{36} + \frac{y^2}{4} = 1$

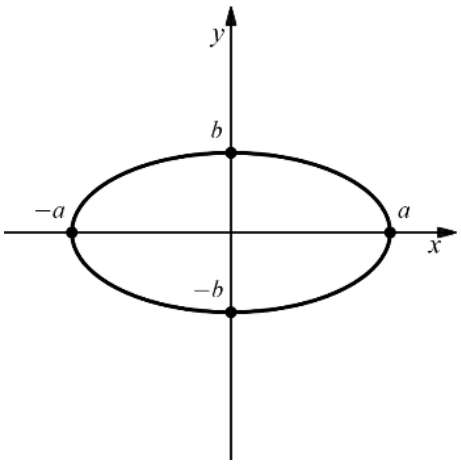


1) e)

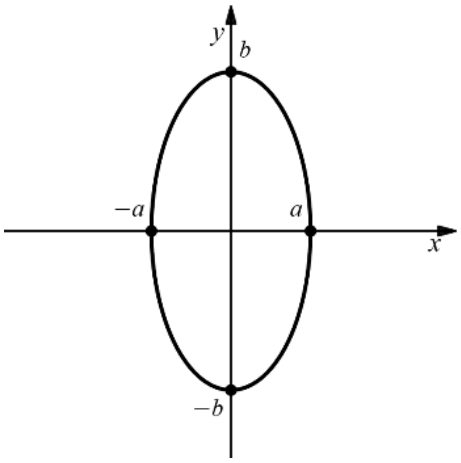
$\frac{x^2}{5} + y^2 = 1$



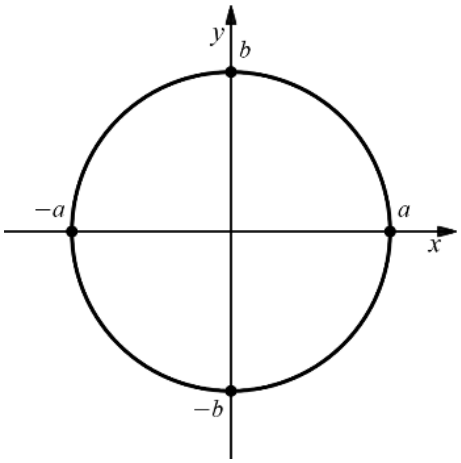
1) f)



1) g)



1) h)



2) a)

$\frac{x^2}{5}+\frac{y^2}{3}=1$ so $(\sqrt{5},0)$ and $(-\sqrt{5},0)$

So $x=-\sqrt{5}$ and $x=\sqrt{5}$

2) b)

$\frac{x^2}{3}+\frac{y^2}{5}=1$ so $(\sqrt{3},0)$ and $(-\sqrt{3},0)$

So $x=-\sqrt{3}$ and $x=\sqrt{3}$

3) a)

$(2,0)$ gives $a=2$

$(0,4)$ gives $b=4$

$a=2$ and $b=4$

3) b)

$(3,0)$ gives $a = 3$

$(0,2)$ gives $b = 2$

$a = 3$ and $b = 2$

3) c)

$(\sqrt{7},0)$ gives $a = \sqrt{7}$

$(0,\sqrt{3})$ gives $b = \sqrt{3}$

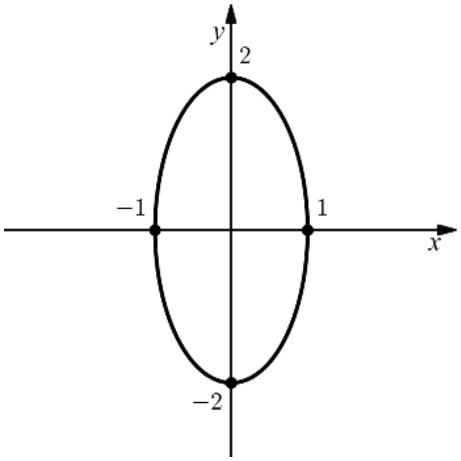
$a = \sqrt{7}$ and $b = \sqrt{3}$

4) a)

$\frac{1}{a} = 1$ and $\frac{4}{b} = 1$

$\therefore a = 1$ and $b = 4$

4) b)

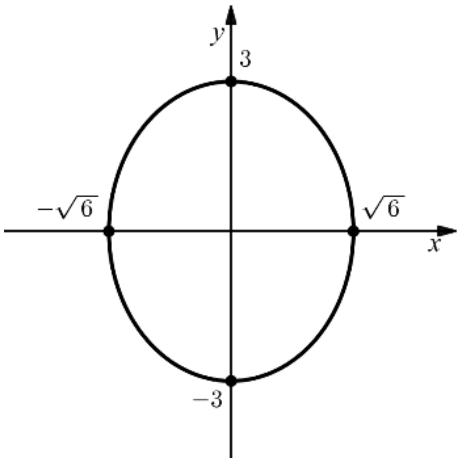


5) a)

$\frac{6}{a} = 1$ and $\frac{9}{b} = 1$

$\therefore a = 6$ and $b = 9$

5) b)



6) a)

$\frac{4}{a}+\frac{6}{b}=1$

$\frac{6}{a}+\frac{3}{b}=1 \Rightarrow \frac{12}{a}+\frac{6}{b}=2$

which gives

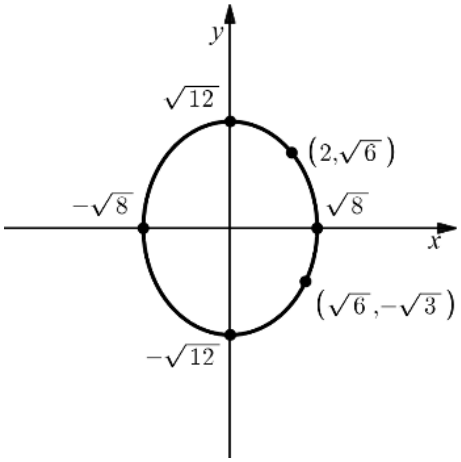
$\frac{4}{a}+\left(2-\frac{12}{a}\right)=1$

$1=\frac{8}{a}$ so $a=8$

$\frac{4}{8}+\frac{6}{b}=1 \Rightarrow \frac{b}{6}=2$

$\therefore a=8$ and $b=12$

6) b)



7) a)

When $y = 0$, $x^2 = a$

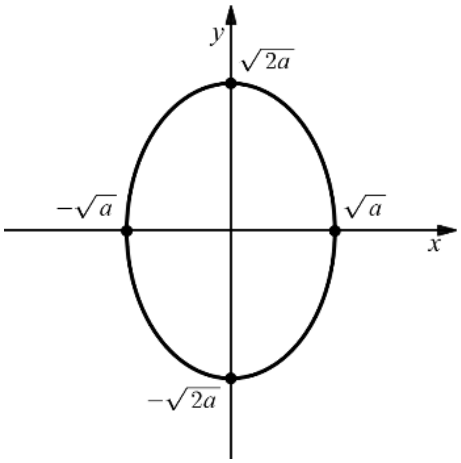
$\therefore \left(-\sqrt{a}, 0\right)$ and $\left(\sqrt{a}, 0\right)$

7) b)

When $x = 0$, $y^2 = 2a$

$\therefore \left(0, -\sqrt{2a}\right)$ and $\left(0, \sqrt{2a}\right)$

7) c)



7) d)

$\frac{4}{a} + \frac{10}{2a} = 1 \Rightarrow \frac{9}{a} = 1$

$\therefore a = 9$